

9.1 $\rightarrow$ Inverse functions9.2 $\rightarrow$ Exponential function.**Meaning Of A Logarithm**

Consider the exponential function $f(x) = 2^x$. Like all exponential functions f is one-to-one. Let's attempt to find a formula for $f^{-1}(x)$:

1. Replace $f(x)$ with y
2. Interchange x and y
3. Solve for Y
4. Replace y with $f^{-1}(x)$

$$y = f(x) \Rightarrow y = 2^x$$

$$x = 2^y$$

$y =$ the exponent to which we raise 2 to get x

$x \rightarrow$ input
 $y \rightarrow$ output

Want to find y for a given x

One-to-one functions always have an inverse.

We now define a new symbol to replace the words "the exponent to which we raise 2 to get x "

$\log_2 x$ is read 'the logarithm, base 2, of x ' or "log, base 2" means "the exponent to which we raise 2 to get x "

Simplifying Logarithms Inverse to $f(x) = a^x$ is $f^{-1}(x) = \log_a x$
($a > 0, a \neq 1$)

1. Simplify $\log_2 32$ by using the definition of a logarithm.

$y = \log_2 32$ means: the exponent to which we raise 2 to get 32 : y such that $2^y = 32$

Therefore, $\log_2 32 = \underline{5}$

2. Simplify $\log_{10} 1000$ by using the definition of a logarithm.

$\log_{10} 1000$ is y for which $10^y = 1000 = 10^3$
 $\Rightarrow y = 3 \Rightarrow \log_{10} 1000 = 3$

3. Simplify $\log_7 49$ by using the definition of a logarithm.

$\log_7 49$ is y for which $7^y = 49 = 7^2$
 $\Rightarrow y = 2 \Rightarrow \log_7 49 = 2$

4. Simplify $\log_8 \frac{1}{8}$ by using the definition of a logarithm.

$\log_8 \frac{1}{8}$ is y for which $8^y = \frac{1}{8} = 8^{-1}$
 $\Rightarrow y = -1 \Rightarrow \log_8 \left(\frac{1}{8}\right) = -1$

5. Simplify $\log_{16} 4$ by using the definition of a logarithm.

$\log_{16} 4$ is y for which $16^y = 4 = 16^{\frac{1}{2}}$
 $\Rightarrow y = \frac{1}{2} \Rightarrow \log_{16} 4 = \frac{1}{2}$

6. Simplify $\log_6 6^5$ by using the definition of a logarithm.

$\log_6 6^5$ is y for which $6^y = 6^5$
 $\Rightarrow y = 5$
 $\Rightarrow \log_6 6^5 = 5$

Exponential functions are one-to-one

$$x = 0 \Rightarrow 2^y = 0 \Rightarrow \text{not possible}$$

$$x = 1 \Rightarrow 2^y = 1 = 2^0 \Rightarrow y = 0$$

$$x = 2 \Rightarrow 2^y = 2 = 2^1 \Rightarrow y = 1$$

$$x = 4 \Rightarrow 2^y = 4 = 2^2 \Rightarrow y = 2$$

y for which $2^y = x$

$x = 1$	$\log_2 1 = 0$
$x = 2$	$\log_2 2 = 1$
$x = 4$	$\log_2 4 = 2$
$\vdots$	$\vdots$
$x = \frac{1}{2}$	$\log_2 \frac{1}{2} = -1$
$x = \frac{1}{4}$	$\log_2 \frac{1}{4} = -2$

Cancellation eqns.: $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$

Example: How would we simplify $3^{\log_3 29}$?

$$3^{\log_3 29} = 29$$

$$a^{\log_a x} = x$$

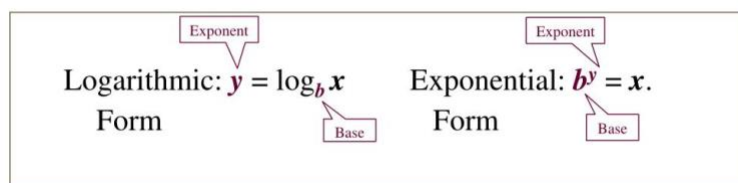
$$\log_a (a^x) = x$$

Definition of a Logarithmic Function

For $x > 0$ and $b > 0, b \neq 1$,

$$y = \log_b x \text{ is equivalent to } b^y = x$$

The function $f(x) = \log_b x$ is the **logarithmic function with base b**.



$$5 = \log_2 32 \quad / \quad 2^5 = 32$$

Note: that logarithm bases must be positive because logarithms are defined as using exponential functions that require positive bases.

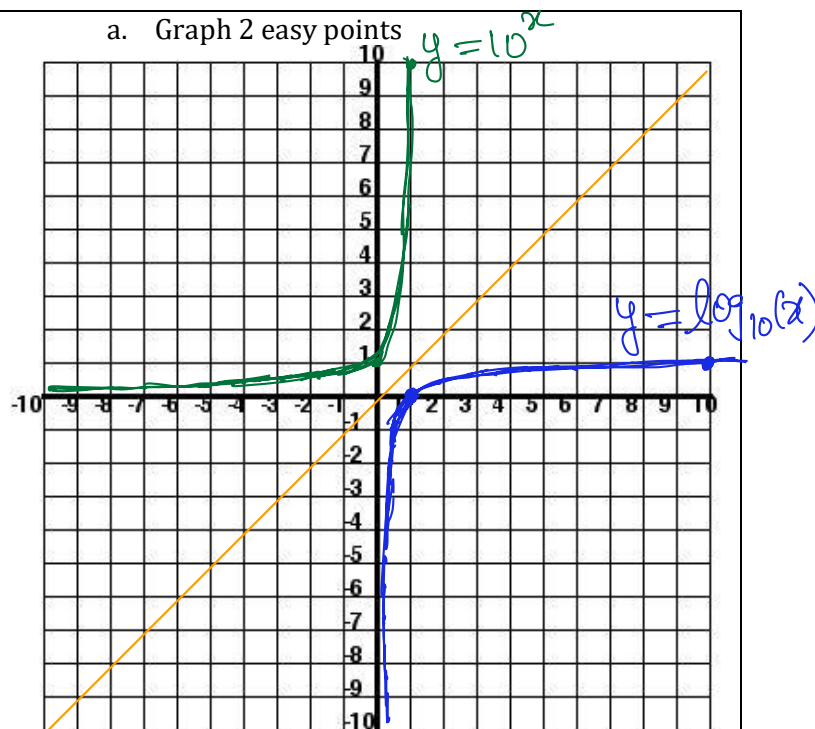
Graphs of Logarithmic Functions

1. Graph $y = \log_{10} x$

a. Create a table using the following x values, then use the rules we learned about simplifying to get the y values

X	Y $= \log_{10}(x)$
$10^{-2} = 0.01$	-2
$10^{-1} = 0.1$	-1
$10^0 = 1$	0
$10^1 = 10$	1
$10^2 = 100$	2

a. Graph 2 easy points

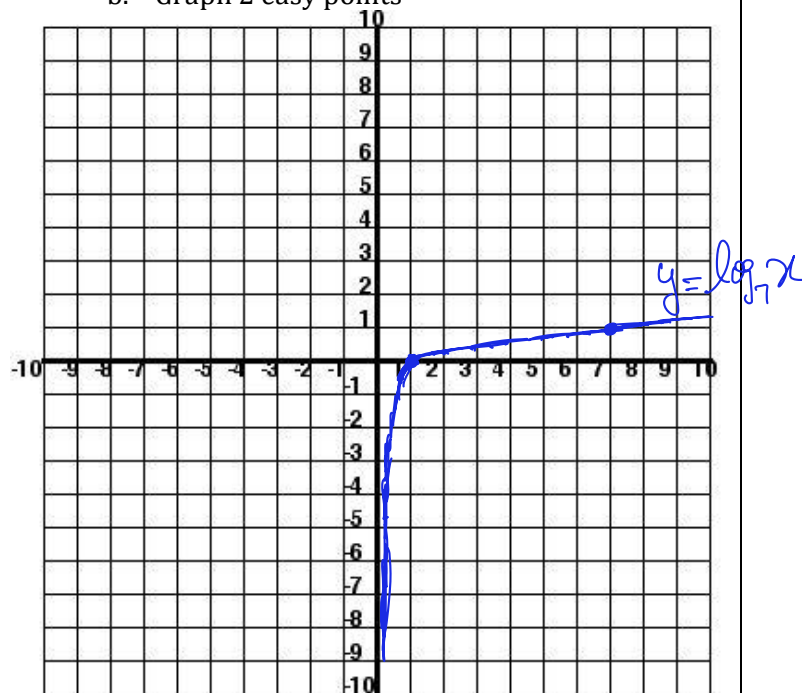


2. Graph $y = \log_7 x$

- b. Create a table using the following x values, then use the rules we learned about simplifying to get the y values

X	Y
$7^{-2} = \frac{1}{49}$	-2
$7^{-1} = \frac{1}{7}$	-1
$7^0 = 1$	0
$7^1 = 7$	1
$7^2 = 49$	2

- b. Graph 2 easy points

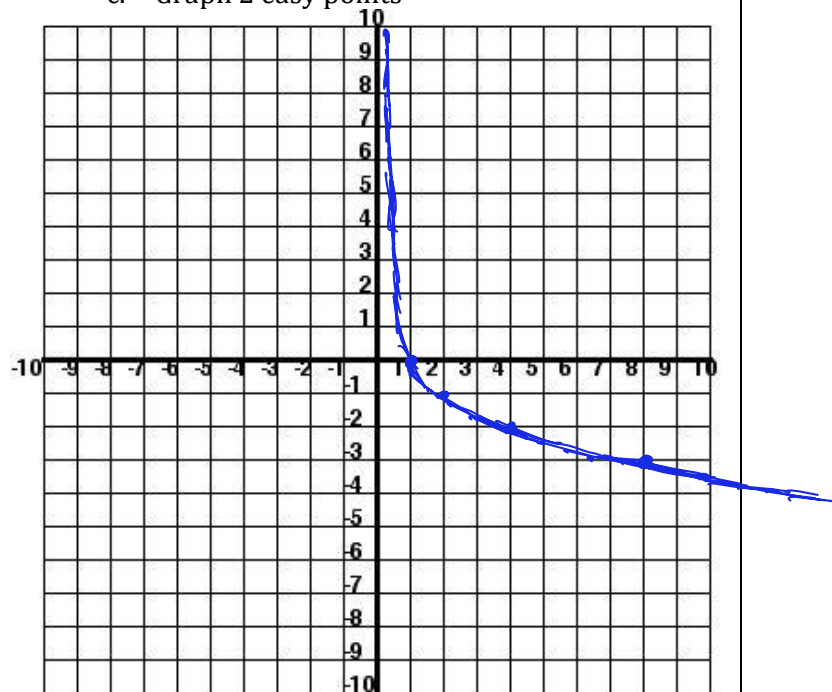


3. Graph $y = \log_{\frac{1}{2}} x$

- c. Create a table using the following x values, then use the rules we learned about simplifying to get the y values

X	Y
$\frac{1}{2}^{-2} = 4$	-2
$\frac{1}{2}^{-1} = 2$	-1
$\frac{1}{2}^0 = 1$	0
$\frac{1}{2}^1 = \frac{1}{2}$	1
$\frac{1}{2}^2 = \frac{1}{4}$	2

- c. Graph 2 easy points



Graphing Exponential and Logarithmic Functions

Graph $f(x) = 3^x$ and $f(x) = \log_3 x$ on the same axis

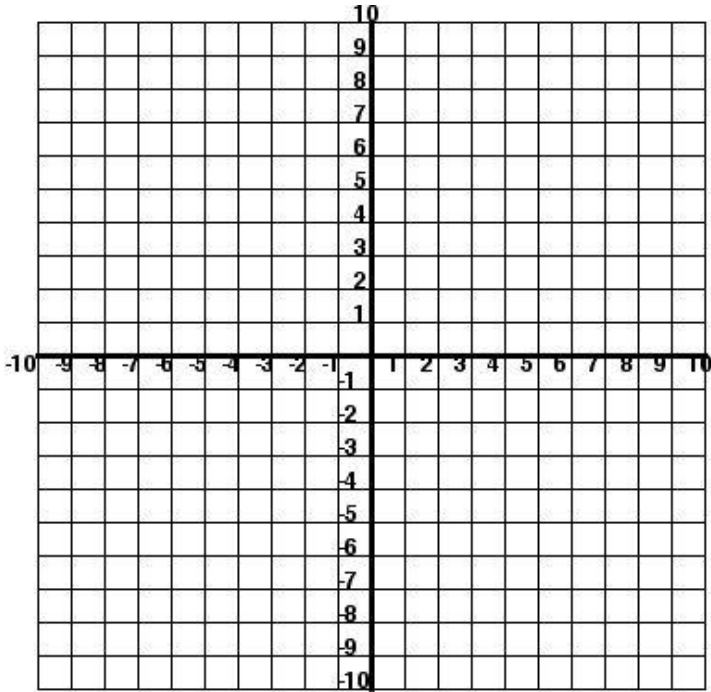
HW. (same as
in $\log_{10} x$)

Table for $f(x) = 3^x$

X	Y
-2	
-1	
0	
1	
2	

Table for $f^{-1}(x) = \log_3 x$

X	Y
___ ⁻² = ___	
___ ⁻¹ = ___	
___ ⁰ = ___	
___ ¹ = ___	
___ ² = ___	



Finding Equivalent Expressions of a Logarithm

Use the definition of a logarithm to rewrite a logarithmic equation as an equivalent exponential equation or the other way around.

$$\log_b(a) = c \iff b^c = a$$

1. $\log_{10} 8 = y$

2. $\log_9 9 = 1$

3. $10^x = 100$

4. $z^m = 6$

Solving Logarithmic Equations

1. Rewrite the logarithm in its exponential form
2. Use the following property to solve $b^m = b^n$ is equivalent to $m = n$

Solve.

1. $\log_6 x = 2$

2. $\log_4 x = 3$

3. $\log_x 9 = 1$

4. $\log_x 7 = \frac{1}{2}$

5. $\log_2 32 = x$

6. $\log_5 25 = x$